

EUROPEAN
CURRICULUM VITAE
FORMAT



PERSONAL INFORMATION:

Name	VIRGOLINI FRANCESCA
Address	VIA C. PERCOTO 10, 33041, AIELLO DEL FRIULI (UD), ITALY
Telephone	(+39) 3394590739
E-mail	francesca.virgolini@gmail.com
Nationality	Italian
Date of Birth	February 5th, 2000
Gender	Female

EDUCATION:

PhD studies

November 2025 – ongoing

Ph.D. candidate in Information Engineering,

Based at the **Intelligent Autonomous Systems Lab**, University of Padua (Italy).

Research topic: Integration of Error-Related Potentials into Shared Intelligence Frameworks for Brain–Machine Interfaces.

Supervisor: Prof. Luca Tonin

Master studies

October 2022 – March 2025

Master of Science in Bioengineering, University of Padova (Padova, Italy).

Master specialization: **Bioengineering for Neuroscience**.

Main courses in Signal Processing, Bioimaging, Machine Learning, Neurorobotics and Neurorehabilitation, Medical Robotics

Title of the final dissertation: ‘Neural Manifolds for Brain-Machine Interfaces: evaluating their potential for Decoding Motor Imagery’. Project carried out at the **Translational Neural Engineering Lab** (TNE Lab) of **EPFL**, Campus Biotech, Switzerland.

Final Score: 110/110 cum laude.

Undergraduate studies

September 2019 – September 2022.

Bachelor’s Degree in Biomedical Engineering, University of Padova (Padova, Italy).

Main courses in Physics, Mathematical Analysis, Calculus, Signals and Systems, Circuit Theory and Electronics, Biomaterials, Biomechanics, Fundamentals of Control Theory, Signal Processing

Title of the final dissertation: ‘Controlling a brain-computer interface: a study on long-term user learning for tetraplegic people’.

Final Score: 110/110 cum laude.

Upper secondary education Diploma at Liceo Scientifico 'Niccolò Copernico' (Udine, Italy).
Curricular optional courses in IT (mainly programming with Java and processing) and practical experiences in biology, geology and chemistry laboratories.

Extracurricular optional courses in Mathematics for Olympic competitions (Combinatorics, Number Theory, Algebra and Geometry) and for National Chess competitions.

Final Examination Score: 100/100.

LABORATORY EXPERIENCES

September 2024 – January 2025

Translational Neural Engineering Lab (TNE Lab), **École polytechnique fédérale de Lausanne (EPFL)**, Campus Biotech, Geneva, Switzerland.

March 2022 – September 2022

Intelligent Autonomous Systems Lab (IAS Lab), University of Padua, Padua, Italy.

WORK EXPERIENCES

2023 - 2024

Teaching Assistant for the course **Elements of Computer Science and Programming** at University of Padova (Padova, Italy). My work consisted in guiding undergraduate students in completing Python coding exercises and in offering assistance with understanding problem-solving approaches, syntax and debugging techniques.

2023 - 2024

Student Orientation Leader, University of Padova (Padova, Italy). My work consists in organizing informational meetings to introduce students to the services and resources available within the university and in addressing student inquiries regarding the curriculum, administrative procedures and other aspects of university life.

2018 - present

Private lessons of Mathematics, Physics, Programming and English to high-school and undergraduate students.

LANGUAGES SPOKEN:

Mother tongue

ITALIAN

Other languages

ENGLISH (Level C1)

Cambridge English Level 3 Certificate (Advanced, C1) – March 2019
Reading, Writing and Speaking: level C1
Use of English: level C2

FRENCH (Level A2)

PROGRAMMING AND IT SKILLS:

Data visualization

MATLAB (Advanced user).

Development of different **projects** for university courses such as: Signals and Systems, Fundamental of Controls Theory, Biomedical Signal Processing, Mechanics for Bioengineering, Medical Informatics (Bachelor's Degree) and Master courses in Biological Signal Processing, Statistical Methods for Bioengineering, Modeling Methodologies for Physiology and Medicine, Neurorobotics and Neurorehabilitation. Noteworthy: Bachelor's Thesis Project.

Programming Languages

Python (Advanced user).

Development of different **projects** for university courses such as Reinforcement Learning and Deep Learning. Teaching Assistant activity for the course of Elements of Computer Science and Programming. Noteworthy: Master Thesis Project.

R (Advanced user).

Development of different projects for university courses such as Machine Learning and Biomarkers, Precision Medicine and Drug Development.

Database Management Systems
(BDMS)

SQL, SQLite (Intermediate user).

Database development for Medical Informatics' course.

DRIVER LICENSE:

B

Date: Padova, 06/11/2025

Signed:

